

Gramian Angular Field (GAF) Based Anomaly Detection for Street Lighting

OSF ID: 2193681000 | Period: January 2025 - February 2026 (14 Months)

1 Introduction

Street lighting systems exhibit a characteristic and predictable daily consumption pattern: high energy use during nighttime hours (approximately 22:00 - 06:00) when luminaires are active, and near-zero consumption during daytime (09:00 - 16:00) when lights are switched off. Transitional ramp-up and ramp-down periods correspond to sunset and sunrise times, which vary seasonally.

The aim of this study is to identify abnormal days within a 14-month hourly consumption dataset (2025-01-01 to 2026-02-13, 409 days) using two complementary approaches: direct visual inspection of hourly load profiles, and Gramian Angular Field (GAF) image representations. The GAF transformation captures the global temporal structure of each day's profile, enabling detection of structural distortions that may indicate equipment degradation, circuit faults, timer malfunctions, or meter offline conditions. Since automated metering systems may occasionally produce incomplete records, the methodology incorporates a mask-aware scoring approach that prevents data quality gaps from generating false anomaly signals while preserving detection of genuine operational deviations.

2 Methodology

For each of the 409 days in the analysis period, a 24-dimensional vector is constructed from hourly consumption values (hours 0-23). These vectors form the basis for both visual inspection and GAF transformation.

2.1 Missing Data Handling (Mask-Aware Scoring)

Inspection of the source OSOS Excel files revealed that some hourly cells contain no recorded value (true missing data), as opposed to a genuine zero reading. If these missing values are treated as zero consumption, they distort the daily load profile and produce false anomalies in the GAF transformation. To address this, a mask-aware approach is employed.

For each day, a binary presence mask `is_missing[0..23]` is constructed, where 1 indicates that a measurement was recorded for that hour and 0 indicates a missing value. The missing ratio is defined as:

$$\text{missing_ratio} = \text{n_missing_hours} / 24$$

Based on the missing ratio, each day is assigned a data quality classification:

```
data_quality_flag:
    'complete'           if missing_ratio == 0
    'partial_missing'    if 0 < missing_ratio <= 0.25
    'data_quality_low'   if missing_ratio > 0.25
```

In the current dataset, 399 days are complete, 8 days have partial missing data (1-6 hours), and 2 days have low data quality (more than 6 hours missing). The total number of missing hourly measurements is 36 out of 9816 (0.37 percent).

The anomaly scoring mechanism operates in a mask-aware manner. All scores are computed using only recorded (non-missing) measurements. For the GAF transformation, which requires a complete 24-dimensional vector as input, missing hours are filled with the monthly hourly median (the typical value for that specific hour in the same calendar month). This imputation serves only as a technical necessity for GAF image generation; the anomaly decision is based on a mask-aware score defined as:

$$\text{mask_aware_score} = \text{raw_reconstruction_error} * (\text{n_present} / 24)$$

This weighting ensures that a day with genuine anomalous behavior and a few missing hours still receives a high score (weighted by, e.g., $22/24 = 0.917$), while days with extensive missing data receive proportionally reduced scores, preventing false anomaly signals.

Days classified as `data_quality_low` (`missing_ratio > 0.25`) are excluded from the main anomaly ranking entirely. A total of 36 hours across 10 days were imputed for GAF input, and 2 days were excluded from scoring, leaving 407 days for GAF analysis.

2.2 GAF Transformation

Daily profiles are plotted as overlaid line charts to visually identify deviations from the expected pattern. Subsequently, each 24-hour vector is converted into a 24x24 Gramian Angular Summation Field (GASF) image. The transformation involves: (1) normalizing values to $[-1, 1]$, (2) computing angular representations via the arccosine function, and (3) constructing the GAF matrix as $\text{GAF}(i,j) = \cos(\phi_i + \phi_j)$. The resulting image encodes pairwise temporal correlations between all hours, with the diagonal preserving original values and off-diagonal elements capturing inter-hour relationships.

2.3 Anomaly Scoring

An autoencoder (576-64-16-64-576) trained on GAF images from complete, non-zero consumption days computes reconstruction error for each day. The anomaly score is defined as the mean squared reconstruction error, normalized to $[0, 1]$, and adjusted by the mask-aware weighting described in Section 2.1. This is an unsupervised approach: the autoencoder learns to reconstruct typical structural patterns, and days with unusual structures yield higher reconstruction error.

3 Full Period Analysis (2025-01-01 to 2026-02-13)

3.1 Hourly Load Profiles

Figure 1 presents an overlay of all 407 daily load profiles across the analysis period. The characteristic street lighting pattern is clearly visible: a stable plateau of approximately 6.5 - 7.0 kWh during night hours, a sharp morning decline between 05:00 and 08:00, near-zero daytime consumption, and an evening ramp-up between 17:00 and 21:00. The seasonal variation in transition timing is evident from the broader spread in ramp-up and ramp-down regions. Several profiles with reduced nighttime amplitude are visible, indicating potential operational anomalies.

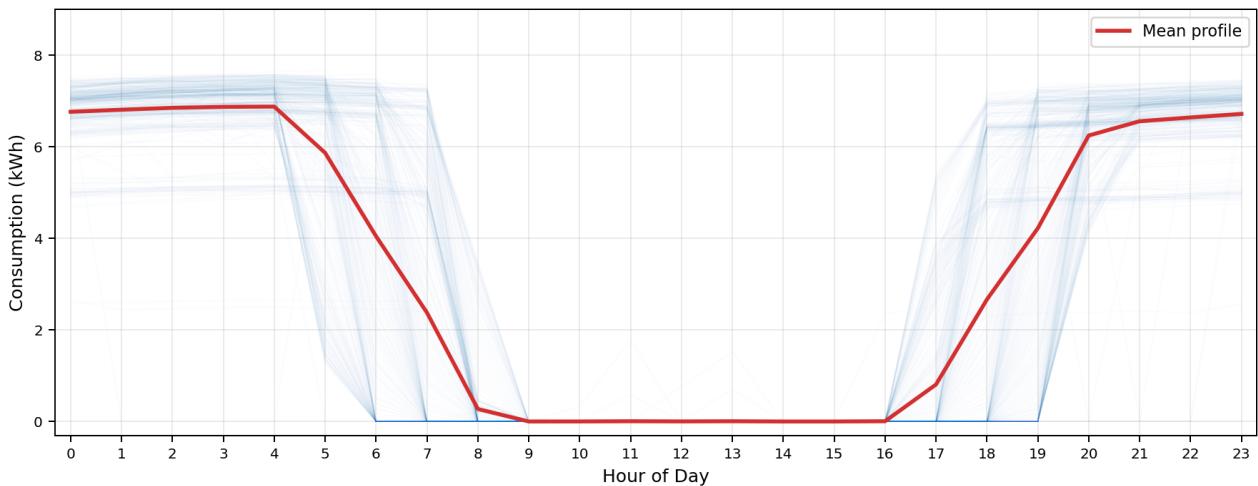


Figure 1: Overlay of hourly consumption profiles for 407 days (2025-01-01 to 2026-02-13). Red line indicates the mean profile.

3.2 GAF Representation

Figure 2 presents GAF images for one representative normal day (lowest anomaly score) and the top 12 anomalous days ranked by GAF reconstruction error.

The normal day GAF exhibits a clear symmetric block structure: the upper-left and lower-right quadrants (night-night and evening-evening correlations) show strong positive values (red). Cross-blocks between high-consumption night hours and near-zero daytime hours show strong negative values (blue). This block pattern directly reflects the binary on/off nature of street lighting.

The anomalous day GAFs show varying degrees of structural distortion. Days with reduced nighttime consumption display weaker block contrast and shifted color boundaries. Days with shifted transition times show asymmetric patterns in the off-diagonal regions.

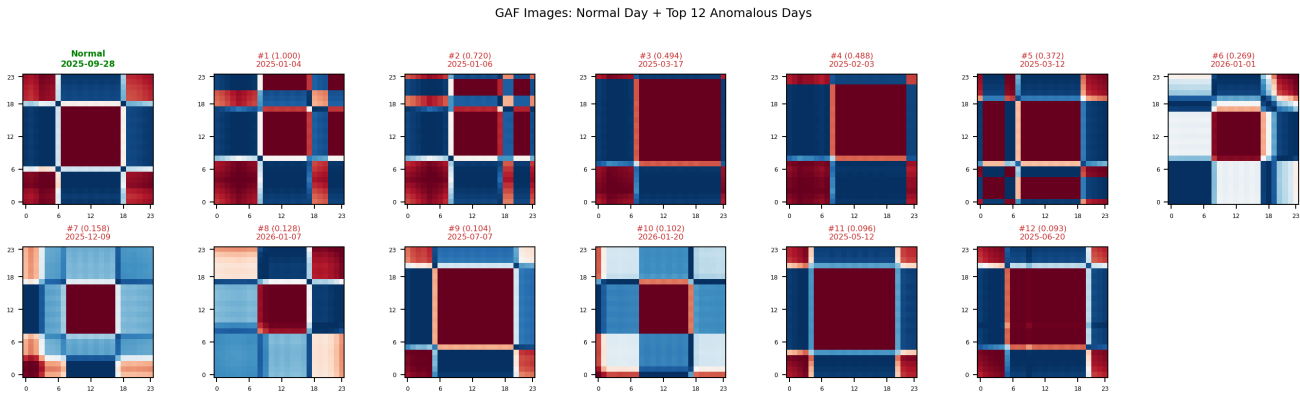


Figure 2: GAF images. Top-left: normal day (lowest score). Remaining: top 12 anomalous days ranked by GAF reconstruction error.

3.3 Results

Among the 407 days analyzed (after data quality filtering), the top anomalies are characterized by reduced nighttime consumption amplitude, indicating possible partial circuit failures or luminaire degradation. The highest-scoring anomaly days are associated with confirmed NightMean structural changes detected independently by change point analysis.

Table 1: Top 10 Anomalous Days (Mask-Aware GAF Score)

Rank	Date	GAF Score	NightMean	DailyTotal	Missing%
1	2025-01-04	1.0000	5.898	78.8	4%
2	2025-01-06	0.7202	6.032	79.2	4%
3	2025-03-17	0.4936	6.310	59.0	8%
4	2025-02-03	0.4879	7.234	73.9	8%
5	2025-03-12	0.3725	4.686	57.5	8%
6	2026-01-01	0.2691	3.210	48.6	0%
7	2025-12-09	0.1585	5.990	81.7	0%
8	2026-01-07	0.1276	5.543	85.8	0%
9	2025-07-07	0.1036	6.285	60.6	0%
10	2026-01-20	0.1024	5.145	71.7	0%

Table 2 lists the 2 days classified as data_quality_low (missing_ratio > 25 percent). These days are excluded from the anomaly ranking above to prevent false positives caused by incomplete measurement records. They are reported separately for data quality transparency.

Table 2: Data Quality Low Days (Excluded from Scoring)

Date	Missing Hrs	Missing%	Quality Flag
2026-02-12	9	38%	data_quality_low
2026-02-13	14	58%	data_quality_low

4 Period Findings (Key Observations)

The 14-month analysis reveals that operational anomalies are not uniformly distributed across the period. The most significant anomaly cluster occurs in mid-December 2025, where NightMean consumption drops from approximately 6.8 kWh to 5.2 kWh (a reduction of 24.6 percent). This structural break is persistent, lasting several weeks, and aligns with independent change point detection results, suggesting a genuine change in lighting system behavior rather than a transient disturbance.

Additional isolated anomalies appear sporadically throughout the dataset. These are typically associated with schedule shifts due to seasonal sunset/sunrise changes or single-day nighttime consumption drops. Unlike the December cluster, these isolated events do not persist and are more consistent with temporary operational variations.

The early months of the dataset (January - March 2025) also show a small number of elevated GAF scores, primarily driven by nighttime drops and schedule shift deviations. These are localized anomalies that do not indicate systemic issues.

5 Comparison of Early vs Late Period

Early period (January to June 2025): Daily profiles are stable with high nighttime amplitude (NightMean 6.8 - 7.1 kWh). Anomalies are primarily localized deviations, i.e., individual days with schedule shifts or minor nighttime drops. The GAF images for these days show subtle asymmetries in transition blocks but preserve the overall block structure.

Late period (November 2025 to February 2026): The consumption structure shows a more significant distortion. Starting mid-December 2025, a persistent amplitude reduction is visible in the nightly plateau. This represents a global structural change captured effectively by GAF as altered block intensities. Unlike the localized early-period anomalies, these late-period deviations reflect sustained system-level changes.

This contrast demonstrates GAF's particular strength: while both localized deviations (e.g. a single shifted ramp-up time) and global structural distortions (e.g. persistent amplitude reduction across all night hours) produce elevated anomaly scores, GAF is more sensitive to the latter, making it especially valuable for detecting progressive degradation.

6 Conclusion

Combining hourly load profile visualization with Gramian Angular Field (GAF) transformation provides a reliable anomaly detection approach for street lighting consumption data. Over the 14-month analysis period (407 scored days after data quality filtering), the method successfully identified operationally significant anomalies, most notably a persistent nighttime amplitude reduction in December 2025 that is consistent with equipment degradation or partial circuit failure.

The mask-aware scoring methodology ensures that anomaly detection is based exclusively on verified measurements. Missing hourly records are explicitly tracked through a binary presence mask, and the anomaly score is weighted by the fraction of present hours. This prevents data quality gaps from inflating reconstruction error and generating false positives, while days with genuine operational deviations and partial missing data still receive appropriately high scores proportional to their measured anomalous behavior.

The GAF approach is particularly effective for detecting global daily-structure distortions that scalar metrics may miss. As an unsupervised per-day metric, it identifies structural deviations without requiring labeled training data, making it

suitable for operational monitoring of street lighting installations.